Festival Shopping Behaviour Analysis

1. Project Overview :

This project analysis customer shopping behaviour using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behaviour to guide strategic business decisions.

2. Dataset Summary :

- Rows: 3,900

- Columns: 19

- Key Features:

- Customer demographics (Customer_ID', 'Age', 'Age_Group', 'Gender', 'Location', 'Subscription_Status')

- Purchase details ('Item_Purchased', 'Category', 'Size', 'Color', 'Season')

- Shopping behavior ('Quantity', 'Purchase_Amount_(USD)', 'Discount_Applied', 'Payment_Method', 'Shipping_Type', 'Review_Rating')

- Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python:

We began with data preparation and cleaning in Python:

● Data Loading: Imported the dataset using pandas.

● Initial Exploration: Used df.info() to check structure and .describe() for summary statistics.

	Customer_ID	Age	Age_Group	Gender	Location	Subscription_Status	Previous_Purchases	Purchase_Frequency_Days	Item_Purchased	Category	Size
0	1	55	Mid-Aged	Male	Kentucky	Yes	14	14	Blouse	Clothing	L
1	2	19	Young Adult	Male	Maine	Yes	2	14	Sweater	Clothing	L
2	3	50	Mid-Aged	Male	Massachusetts	Yes	23	7	Jeans	Clothing	S
3	4	21	Young Adult	Male	Rhode Island	Yes	49	7	Sandals	Footwear	M
4	5	45	Mid-Aged	Male	Oregon	Yes	31	365	Blouse	Clothing	M

Color	Season	Quantity	Purchase_Amount_(USD)	Discount_Applied	Payment_Method	Shipping_Type	Review_Rating
Gray	Winter	4	53	Yes	Venmo	Express	3.1
Maroon	Winter	2	64	Yes	Cash	Express	3.1
Maroon	Spring	1	73	Yes	Credit Card	Free Shipping	3.1
Maroon	Spring	4	90	Yes	PayPal	Next Day Air	3.5
Turquoise	Spring	3	49	Yes	PayPal	Free Shipping	2.7

● Missing Data Handling: Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

● Column Standardization: Renamed columns to snake case for better readability and documentation.

● Feature Engineering: ○ Created age_group column by binning customer ages. ○ Created purchase_frequency_days column from purchase data.

● Data Consistency Check: Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.

● Database Integration: Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions):

We performed structured analysis in MySQL to answer key business questions:

1. Revenue by Gender – Compared total revenue generated by male vs. female customers.

Gender	revenue
Male	157890
Female	75191

2. Top 5 Products by Rating – Found products with the highest average review ratings.

Item_Purchased	avg_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Skirt	3.79

3. High-Spending Discount Users – Identified customers who used discounts but still spent above the average purchase amount.

Customer_ID	Purchase_Amount_(USD)
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88
29	94
32	79

4. Shipping Type Comparison – Compared average purchase amounts between Standard and Express shipping.

Shipping_Type	avg_purchase
Express	60.48
Standard	58.46

5. Subscribers vs. Non-Subscribers – Compared average spend and total revenue across subscription status.

Subscription_Status	total_customers	avg_spend	total_revenue
No	2847	59.87	170436
Yes	1053	59.49	62645

6. Discount-Dependent Products – Identified 5 products with the highest percentage of discounted purchases.

Item_Purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

7. Customer Segmentation – Classified customers into New, Returning, and Loyal segments based on purchase history.

customer_segment	number_of_customers
Loyal	3116
Returning	701
New	83

8. Top 3 Products per Category – Listed the most purchased products within each category.

Category	Item_Purchased	total_orders
Accessories	Jewelry	171
Accessories	Sunglasses	161
Accessories	Belt	161
Clothing	Blouse	171
Clothing	Pants	171
Clothing	Shirt	169
Footwear	Sandals	160
Footwear	Shoes	150
Footwear	Sneakers	145
Outerwear	Jacket	163
Outerwear	Coat	161

9. Repeat Buyers & Subscriptions – Checked whether customers with >5 purchases are more likely to subscribe.

Subscription_Status	repeat_buyers
Yes	958
No	2518

10. Revenue by Age Group – Calculated total revenue contribution of each age group.

Age_Group	total_revenue
Mid-Aged	89741
Adult	67188
Senior	46894
Young Adult	29258

5. Dashboard in Power BI :

Finally, we built an interactive dashboard in Power BI to present insights visually.

6. Business Recommendations:

- Boost Subscriptions – Promote exclusive benefits for subscribers.
- Customer Loyalty Programs – Reward repeat buyers to move them into the "Loyal" segment.
- Review Discount Policy – Balance sales boosts with margin control.
- Product Positioning – Highlight top-rated and best-selling products in campaigns.
- Payment Method Partnerships – Collaborate with popular digital payment providers for festive cashback offers.
- Targeted Marketing – Focus efforts on high-revenue age groups and express-shipping users.
- Customer Feedback Utilization – Leverage review ratings to improve product quality and trust.